卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-ENG-ECI-DD-GL-0103 30-B003701S-Z0103

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

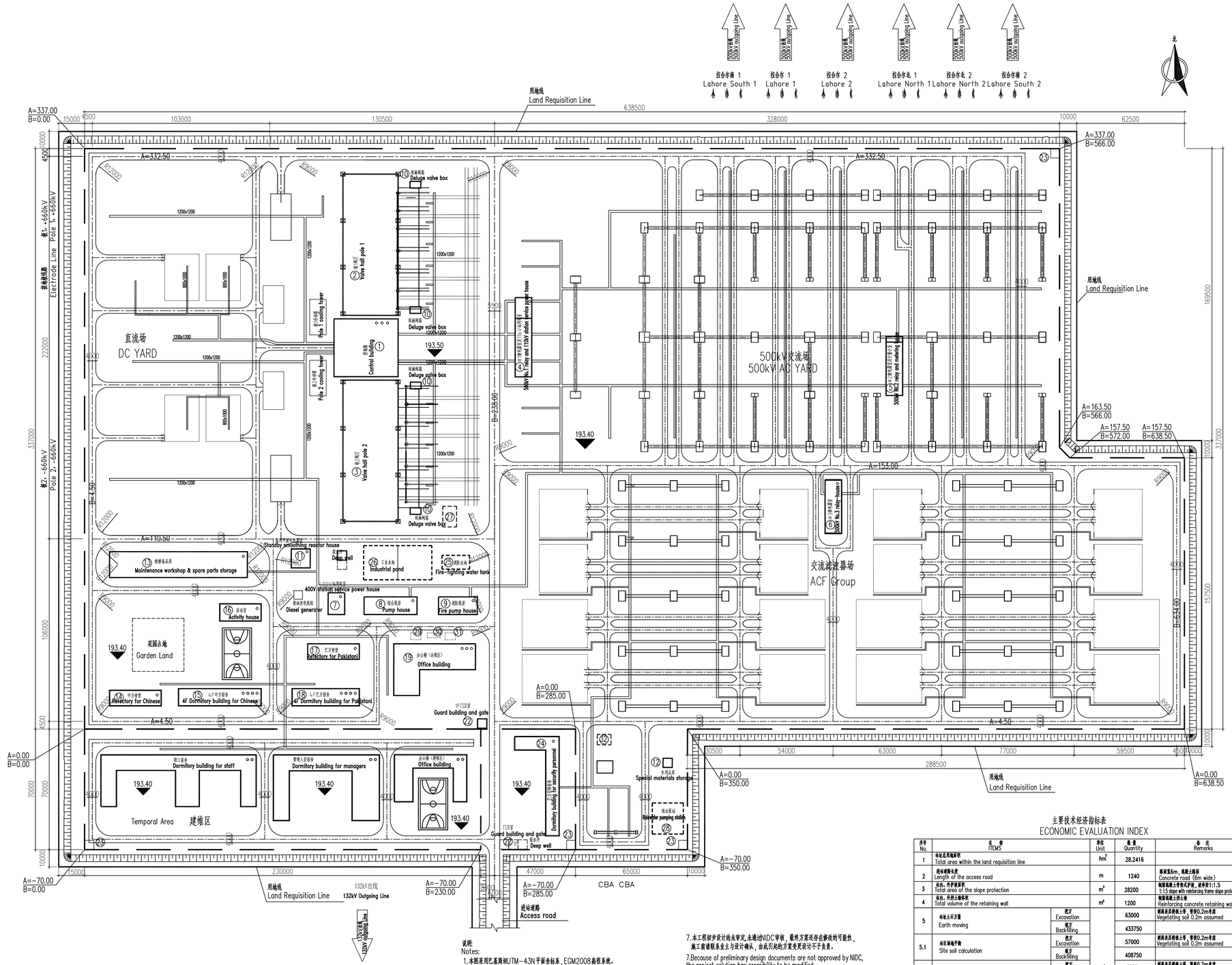
总图 专业 第 01 卷 第 03 册
General Layout SPECIALITY VOLUME BOOK

卷册名称 场地初平图
Volume name Site Leveling Diagram

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	宋宾鹏
图 纸 3 张 DRAWING 3 NOS.	打印件 0 本 PRINTED COPIES 0 NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	B003701S-Z0103-01	场地初平图 Site Leveling Diagram	A	1		
2	B003701S-Z0103-02	竖向布置图 Vertical Layout Diagram	A	1		
2	B003701S-Z0103-03	土方平衡图 Earthwork Balance Diagram	A	1		

注:



场地初平图

Field Leveling Diagram

说明:

Notes:

1.本图采用巴基斯坦UTM-43N平面坐标系，EGM2008高程系统.

1.Coordinate system: Pakistan UTM-43N. Elevation system: EGM2008

2.相对平面坐标以米(m)为单位，尺寸以毫米(mm)为单位.

2.Relative coordinate systems is measured in meter. Dimensions are measured in millimeters.

3.场地平均设计标高为193.40m.

3.Site level is equal to 193.40m.

4.场内道路为郊区型道路，道路中心标高193.60m.

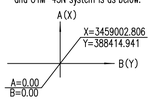
4.Interior Road is suburban type, the center level of interior road is 193.60m

5.未注明的道路宽度均为3.0米，道路的转弯半径一般均为7.0米.

5.The width of interior roads which are not indicated is 3.0m. The average turning radius of interior roads is 7.0m

6.本图建筑(A、B)坐标与UTM-43N平面坐标系关系如下所示:

6.The relationship between relative coordinate system and UTM-43N system is as below:



7.本工程初步设计尚未审定，未通过NIDC审核，最终方案还存在修改的可能性，施工前请联系业主与设计确认，由此引起的方案变更设计不予负责.

7.Because of preliminary design documents are not approved by NIDC, the project solution has possibility to be modified. The design contractor has no responsibility for the project solution changed, if the construction contractor not get authorized by project owner and design contractor before construction started.

8.根据初设原则审查会沟通结论，本站西侧按围墙外新增5m进行站区规平，围墙位置暂保持不变，最终方案待初设审定后修改.

8.According to the conclusion of principle examination meeting for preliminary design, site level work is formed as 5m away from west side of enclosing wall, the position of enclosing wall temporarily stay unchanged, the final scheme is pending for final approval of preliminary design.

9.本图以CP-3、CP-10、CP-6、CP-8为定位点，坐标分别为:

9.The drawing uses point CP-3, CP-10, CP-6 and CP-8 as locating points, and the corresponding coordinates are as below:

CP-3	X(m)=3458638.678	CP-10	X(m)=3458726.757
191.166	Y(m)=388554.185	191.521	Y(m)=389586.869
CP-6	X(m)=3459413.990	CP-8	X(m)=3459420.705
191.354	Y(m)=389040.061	191.066	Y(m)=388398.196

10.进场道路宽度6.0m，自站区自侧接入Bhai Pheru-Mor Khunda Road公路，长度约1240m.

10.The width of access road is 6.0m. The access road connects to the Bhai Pheru-Mor Khunda Road highway. And the length of it is about 1240m.

建(构)筑物一览表

Buildings and structures list

编号 No.	名 称 Name of buildings	建筑面积(㎡) Architectural area(㎡)	占地面积(㎡) Area covered(㎡)	备 注 Remarks
①	控制楼 Control building	3243	1309	三层框架结构 Reinforcing frame structure (3 stories)
②	阀厅1柱厅 Valve hall pole 1	3075	3075	钢结构(单层) Steel structure (single story)
③	阀厅2柱厅 Valve hall pole 2	3075	3075	钢结构(单层) Steel structure (single story)
④	多1柱1号变压器110/4/6用电室 500kV No.1 relay and 110kV station service power house	455	455	单层框架结构 Reinforcing frame structure (single story)
⑤	多2柱电量变压器计量小室 500kV No.2 relay and metering house	341	341	单层框架结构 Reinforcing frame structure (single story)
⑥	多3柱电量室 500kV No.3 relay-house	308	308	单层框架结构 Reinforcing frame structure (single story)
⑦	400V站用电室 400V station service power house	126	126	单层框架结构 Reinforcing frame structure (single story)
⑧	综合泵房 Pump house	315	315	单层框架结构 Reinforcing frame structure (single story)
⑨	消防泵房 Fire pump house	231	231	单层框架结构 Reinforcing frame structure (single story)
⑩	雨淋阀箱 Deluge valve box		24x4	成品 Finished products
⑪	备用平波电抗器室 Standby smoothing reactor house		80	钢结构(单层) Steel structure (single story)
⑫	专用品库 Special materials storage	28.5	28.5	单层框架结构 Reinforcing frame structure (single story)
⑬	维修备品库 Maintenance workshop & spare parts storage	1274	1274	单层框架结构 Reinforcing frame structure (single story)
⑭	中方食堂 Refectory for Chinese	280	280	单层框架结构 Reinforcing frame structure (single story)
⑮	中方宿舍 Dormitory building for Chinese	1888	472	四层框架结构 Four storied framed structure
⑯	活动室 Actively house	231	231	单层框架结构 Single framed structure
⑰	巴方食堂 Refectory for Pakistani	280	280	单层框架结构 Single framed structure
⑱	巴方宿舍 Dormitory building for Pakistani	1528	383	四层框架结构 Reinforcing frame structure (4 stories)
⑲	办公楼(运维区) Office building (Operation and Maintenance Area)	2052	678	三层框架结构 Reinforcing frame structure (3 stories)
⑲a	办公楼(建维区) Office building (Temporal Area)	1424	712	二层砖砌结构 Masonry structure (2 stories)
⑳	管理人员宿舍 Dormitory building for managers	2357	1178	二层砖砌结构 Masonry structure (2 stories)
㉑	职工宿舍 Dormitory building for staff	2784	1518	二层砖砌结构 Masonry structure (2 stories)
㉒	警卫室 Guard building and gate	32x2	32x2	单层砖砌结构 Masonry structure (single stories)
㉓	瞭望塔 Watchtower	22x4	22x4	二层框架结构 Reinforcing frame structure (2 stories)
㉔	巴方安保宿舍 Dormitory building for security personnel	697	697	单层砖砌结构 Masonry structure (single stories)
㉕	消防水池 Fire-fighting water tank		128	地下钢筋混凝土结构 Underground reinforcing concrete structure
㉖	工业水池 Industrial pond		784	地下钢筋混凝土结构 Underground reinforcing concrete structure
㉗	油污水分离池 Oily water seperate tank for covert transformer		96	地下钢筋混凝土结构 Underground reinforcing concrete structure
㉘	雨水泵站 Rainwater pumping room		211	地下钢筋混凝土结构 Underground reinforcing concrete structure
㉙	污水池 Sewage pit		24	地下钢筋混凝土结构 Underground reinforcing concrete structure
㉚	污水处理装置 Sewage treatment device		21	地下钢筋混凝土结构 Underground reinforcing concrete structure
㉛	污水回收池 Sewage recycling pit		24	地下钢筋混凝土结构 Underground reinforcing concrete structure
㉜	备用变油污水池 Oily water seperate tank for auxiliary transformer		47	地下钢筋混凝土结构 Underground reinforcing concrete structure
合计 Total		26145	18630	

图例:

Legend:

建筑物 Building
道路 Road
围墙 Enclosing wall
地下建筑 Building underground
架构 Substaion frame
电缆沟 Cable trench

主要技术经济指标表

ECONOMIC EVALUATION INDEX

序号 No.	名 称 ITEMS		单位 Unit	数量 Quantity	备 注 Remarks
1	场址范围征地面积 Total area within the land requisition line		hm^2	28.2416	
2	进场道路长度 Length of the access road		m	1240	路面宽6.0m，双向三车道 Concrete road (6m wide)
3	场内、外护坡面积 Total area of the slope protection		m^2	28200	砌筑混凝土骨架植草护坡，坡率取1:1.5 1:1.5 slope with reinforcing frame slope protection
4	场内、外挡土墙体积 Total volume of the retaining wall		m^3	1200	钢筋混凝土挡土墙 Reinforcing concrete retaining wall
5	场址土石方量 Earth moving	挖方 Excavation	m^3	63000	铺植表层耕植土平均，管铺0.2m考虑 Vegetating soil 0.2m assumed
		填方 Backfilling		433750	
5.1	场区场地平衡 Site soil calculation	挖方 Excavation		57000	铺植表层耕植土平均，管铺0.2m考虑 Vegetating soil 0.2m assumed
		填方 Backfilling		408750	
5.2	进场道路 Access road	挖方 Excavation		6000	铺植表层耕植土平均，管铺0.2m考虑 Vegetating soil 0.2m assumed
		填方 Backfilling		25000	
5.3	场址土方综合平衡后情况 Site Earthwork Comprehensive Balance	弃土 Abandoned		42000	
		借土 Buying		512800	外购土方 Buying
6	场内道路面积 Interior Road		m^2	29860	混凝土路面 Concrete road
7	主电缆沟长度 Cable trenches		m	6700	0.6mX0.6m及以上 Section size over 0.6mX0.6m
8	总建筑面积 Total architectural area		m^2	26145	
9	屋内配电装置场地铺装面积 Site pavement		m^2	105000	土工布，铺150mm碎石 Geotextile and 150mm gravel pavement
10	新进变配电站轨道长度 Rail track		m	2300	

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT & TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - E N G - E C I - C D - G L - 0 1 0 3 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	荣爱卿	
校核 Checked	刘俨	
审核 Reviewed	毛洁清/李洁	
批准 Approved	王子岳	
比例 Scale		

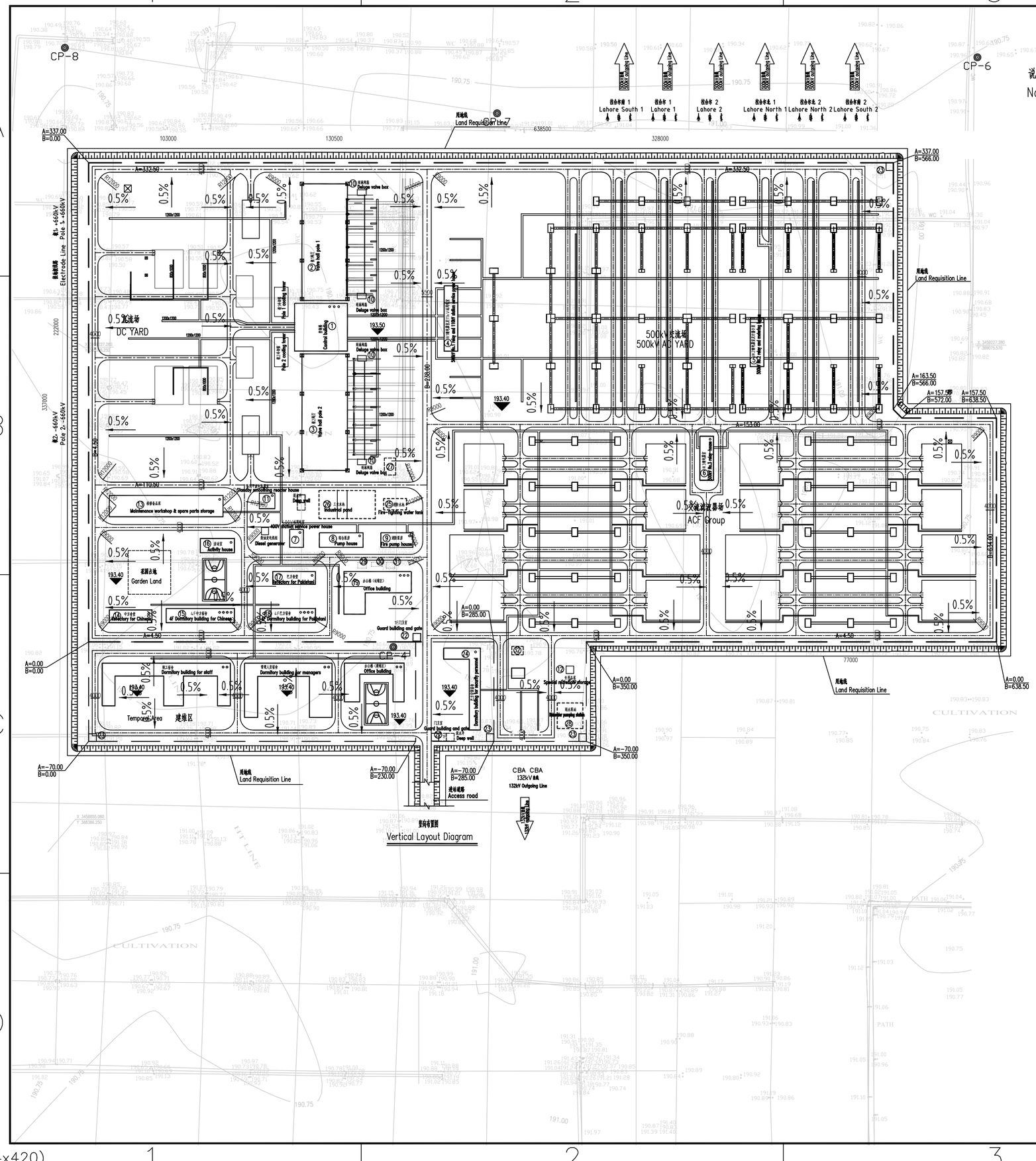
场地初平图

Site Leveling Diagram

图号 DWG NO. B003701S-Z0103-01

版本 REV. A

A1(841x594)



说明:

Notes:

1.本图是根据我院2015年10月提供的测量图设计绘制。

1.This drawing is based on survey graph OCT, 2015.

采用巴基斯坦UTM-43N平面坐标系，EGM2008高程系统。

Coordinate system: Pakistan UTM-43N. Elevation system: EGM2008

2.平面坐标以米为单位，尺寸以毫米为单位。

2.Plane coordinates system is measured in meter. Dimensions are measured in millimeters.

3.场地平均设计标高暂定为193.40m。

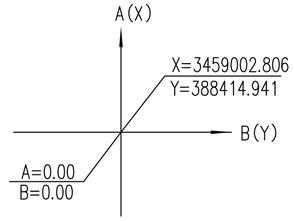
3.Site level is equal to 193.40m.

4.竖向布置采用平坡式布置。

4. Flat slope is applied in vertical layout.

5.本图建筑（A、B）坐标与UTM-43N平面坐标系关系如下所示:

5.The relationship between relative coordinate system and UTM-43N system is as below:



6.场地排水采用有组织排水，场地排水坡度0.5%。坡向雨水井，集中流向雨水泵站。

电缆沟与道路周围的封闭场地设置雨水井，以利排水。

6. Organized sewage is applied in site, the drainage slope is 0.5%. Slope aspect face towards rainwater well, rainwater is concentrated and flows to rainwater pumping station. Located rainwater well at the enclosed area around cable trench and road for the benefit of sewage.

7.站外排水方案尚未确定。

7. Sewage scheme outside station is not confirmed yet.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - E N G - E C I - D D - G L - 0 1 0 3 - 0 0 2 A

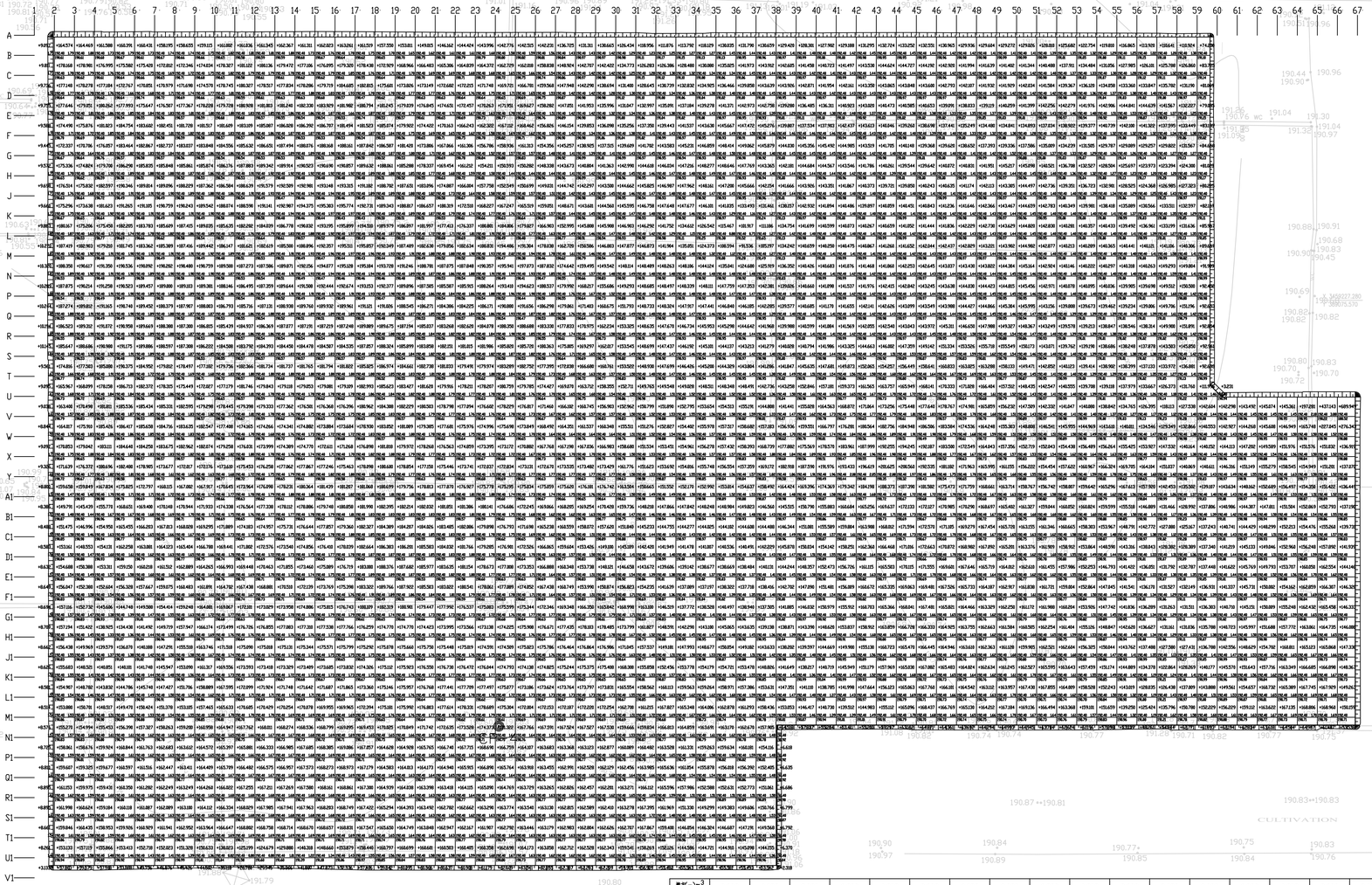
	姓名 Name	日期 Date
设计 Designed	宋宏福	
校核 Checked	刘艳	
审核 Reviewed	韩建强/李静	
批准 Approved	张志强	
比例 Scale		图号 DWG NO. B003701S-Z0103-02

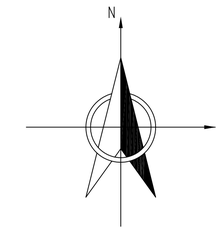
竖向布置图

Vertical Layout Diagram

版本 REV. A

A2(594x420)





说明：

Notes：

1.本图根据我院2015年10月提供的测量图设计绘制。

1.This drawing is based on measurement survey OCT, 2015.

采用巴基斯坦UTM-43N平面坐标系，EGM2008高程系统。

Coordinate system: Pakistan UTM-43N. Elevation system: EGM2008

2.换流变区域平均设计标高取193.40m；建缩区场地平均设计标高取193.40m。

2.Site level is equal to 193.40m.

3.场地平整方格网尺寸为10mx10m。

3.Site level grid dimension is 10mx10m.

施工高度(m) Constructional elevation(m)	初平标高192.4(m) Initial elevation 192.4(m)
	自然高度(m) Natural elevation (m)

4.网格计算图中未包括建构筑物基础、道路、电缆沟的施工余土以及耕植土、淤泥。

4.Grid Calculation drawing excludes the base of buildings, roads, the residual soil during constructing cable trench, planting soil and sludge.

5.全站土方综合平衡工程量如下表：

5.The engineering quantity of site earthwork comprehensive balance is considered as following table:

初平土方平衡表
Initial earth moving table

类 别 Type	挖方(m^3) Excavation	填方(m^3) Backfilling	备 注 Remarks
站区场地 Substation field	0	385750	网格计算 Grid calculation
进站道路 Access road	6000	25000	
考虑大型机械进场及软弱土排水固结 Consider the entry of Large machinery and drainage consolidation of soft soil		23000	压实沉陷均0.1m Grid calculation
挖除表层耕植土 Stripping top planting soil	57000		剥离表层耕植土等，暂按0.2m考虑，包含站区及进站道路。 Take 0.2m for stripped top planting soil etc in to account, including substation area and access road
合计 Total	63000	433750	

考虑基础及沟道余土均：182000m^3

Consider base and residual soil of channels approximately: 182000m^3

考虑是外配电装置场地处理碎石约：16000m^3

Consider to cover chipping stone on switchgear yard approximately: 16000m^3

本工程受百年一遇洪水位影响，换流变区域终平标高取193.40m；建缩区终平标高取193.40m。

Consider the influence of 100-year flood water level, substation/construction area final elevation is 193.40m

则须外取填土：(193.4-192.4)x235054+433750+63000-182000-16000=533800m^3

So that external filling soil is required: (193.4-192.4)x235054+433750+63000-182000-16000=533800m^3

综上，站址土方综合平衡后需：

In conclusion, after site earthwork comprehensive balance:

弃土：42000m^3(剥离表层耕植土等，暂按0.2m考虑，晒干去除植物根至回填场地21000方)

Abandon soil: 42000m^3(Stripped top planting soil 0.2m, dry removal of plant roots backfill site 21000m^3)

购土：512800m^3(考虑耕植土21000方返还利用)

Phurchase soil: 512800m^3(21000m^3 plant soil is considered as return utilization)

8.场地平整要求：场地内的建筑垃圾应予清除干净。场地内填方应分层夯实，压实系数不小于0.95。

8. Requirement for leveling: Constructional waste in site shall be cleaned up.Site fill shall be compacted layer by layer. Compaction coefficient should no less than 0.95.

9.填料要求：

9.Requirement for backfilling material:

1)级配良好的碎石土。

1) Well graded gravel soil.

2)以砾石、卵石或块石作填料时，分层夯实时其最大粒径不宜大于300mm，分层压实时其最大粒径不宜大于100mm。

2) In the case of using gravel, pebble or stone block as filling material, the maximum particle diameter should no greater than 300mm during layered compact. the maximum particle diameter should no greater than 100mm during layered compression.

3)以粉质粘土、粉土作填料时，其含水量宜为最佳含水量，可采用击实试验确定。

3) In the case of using silty clay, silt as filling material, it is preferable to use optimum moisture content, and use compaction test to confirm.

4)不得使用淤泥、耕土、冻土、膨胀性土以及有机质含量大于5%的土。

4) Sludge, planting soil, frozen soil, expansive soil and soil contains greater than 5% organic matter is forbidden.

10.填土施工的要求：

10.Requirement for backfilling construction:

1)应先挖除耕植土及河塘淤泥，部分耕植土需返还作为绿化表土用。

1) Plant soil and pond sludge should be stripped first, part of plant soil will be returned as green surface.

2)分层填料的厚度、分层压实的遍数应根据所选用的压实设备并通过试验确定。

在雨季、冬季进行压实填土施工时，应采取防雨、防冻措施。

2) Thickness of layered compact, frequency of layered compression should be determined by compression equipment and experiment. During compression in rainy season and winter, measures should be taken for rain and frozen protection.

防止填料(粉质粘土、粉土)受雨水淋湿或冻结，并应采取措施防止出现橡皮土。

To prevent filling material (silty clay, silt) being rained out or frozen. Measures should be taken to prevent rubber soil.

3)填土的施工缝各层应错开搭接，在施工缝的搭接处，应适当增加压实遍数。

3) The constructional filling joint should be overlapped in each layer. Number of times of compression should be increased appropriately in the joints.

4)场地回填土压实系数大于0.95。

4) Compaction coefficient should greater than 0.95.

11.场地平整土石方施工和质量检测应符合美国现行规程规范的有关规定。

11. Field leveling earthwork construction and quality examination should satisfy relevant existing US regulations.

12.施工单位在场地平整前，需核实方格网内的自然标高、土方量以及淤泥厚度，并由监理单位确认，如与设计有出入，请与设计院联系，以避免产生较大误差。

12.Before leveling, construction company should exam the natural elevation, earthwork and thickness of sludge in grid,and confirmed with field supervisor company. If divergence is detected, please contact the designer to prevent further deviation.

| | | |